

11

Atom interferometry

11.1 Young's double-slit experiment	247
11.2 A diffraction grating for atoms	249
11.3 The three-grating interferometer	251
11.4 Measurement of rotation	251
11.5 The diffraction of atoms by light	253
11.6 Conclusions	257
Further reading	258
Exercises	258

The possibility of interferometry with atoms follows directly from wave–particle duality. The wave-like propagation of particles, such as atoms, means that they undergo interference and diffraction in an analogous way to light. This chapter explains how such matter waves have been used in interferometers that measure rotation and gravitational acceleration to a precision comparable with the best optical instruments. As in many important developments in physics, atom interferometry relies on simple principles—the first part of this chapter uses only elementary optics and the de Broglie relation

$$\lambda_{\text{dB}} = \frac{h}{p} \quad (11.1)$$

for the wavelength of the matter wave associated with a particle of momentum $p = Mv$. This relation between wavelength and momentum also applies to light waves and the momentum of photons, but here we shall use λ_{dB} exclusively for matter waves and λ , without a subscript, for the wavelength of light.

A sodium atom with velocity $v = 1000 \text{ m s}^{-1}$ (typical of a thermal beam) has $\lambda_{\text{dB}} = 2 \times 10^{-11} \text{ m}$ —about 1/30 000 times the wavelength of visible light, and comparable to that of X-ray radiation. Gratings with lines sufficiently close together to diffract such short wavelengths can now be made by nano-fabrication, i.e. techniques for making structures on scales smaller than $1 \mu\text{m}$.¹ Neutrons also have short de Broglie wavelengths but, unlike atoms, they pass through crystals and diffract from the closely-spaced planes of atoms. Electrons also diffract from crystals and their wave properties were observed long ago by Davisson and Germer in their classic experimental confirmation of wave–particle duality in quantum physics. This previous work with neutrons and electrons is mentioned here simply to show that matter-wave interferometry has a long history, and both neutron and electron diffraction are now highly-developed techniques used in condensed matter physics (Blundell 2001). The more recent matter-wave experiments with atoms described in this chapter should not be regarded as tests of already well-known quantum behaviour; rather their importance lies in their ability to make more precise measurements than other techniques in certain applications.

Consideration of the familiar Young's double-slit experiment provides a good introduction to the basic ideas of matter-wave experiments with atoms and gives a feeling for the size of the physical parameters. We shall then extend the treatment to a diffraction grating (multiple slits) and the design of an interferometer that measures the rate of rotation by

¹It might seem advantageous to use laser-cooled atoms with larger λ_{dB} , but we shall see that this is not true.

the Sagnac effect. These matter-wave experiments operate on the same principles as similar experiments with light because the atoms remain in the ground state and propagate as simple waves. It will be assumed that the reader understands the standard treatment of Fraunhofer diffraction in optics (Brooker 2003) and key results will be quoted here rather than derived. After the discussion of the work with nano-fabricated slits and gratings, we shall look at the use of laser light to manipulate the atom's momentum using methods closely related to laser cooling. These laser techniques make use of the atom's internal energy levels—something that is not possible for electrons or neutrons.

11.1 Young's double-slit experiment

Young originally carried out his double-slit experiment to test the wave nature of light and his simple arrangement still finds practical use in measurements of the coherence of light.² Figure 11.1 shows a typical experimental layout. Waves propagate from the source slit S through the two slits, Σ_1 and Σ_2 , to a point P in the detection plane.³ The amplitude of the light at any point on the detection plane equals the sum of the electric field amplitudes that arrive at that point via slits Σ_1 and Σ_2 . In any interference or diffraction calculation the resultant amplitude at the final point is determined by summing the contributions from all possible paths *taking account of the phase*. For the double slits

²The double slits also form the basis of many theoretical discussions of fundamental issues in quantum mechanics, such as why we cannot know which slit the photon went through and still observe interference.

³Young's fringes are not localised on this plane but can be seen throughout the far-field region.

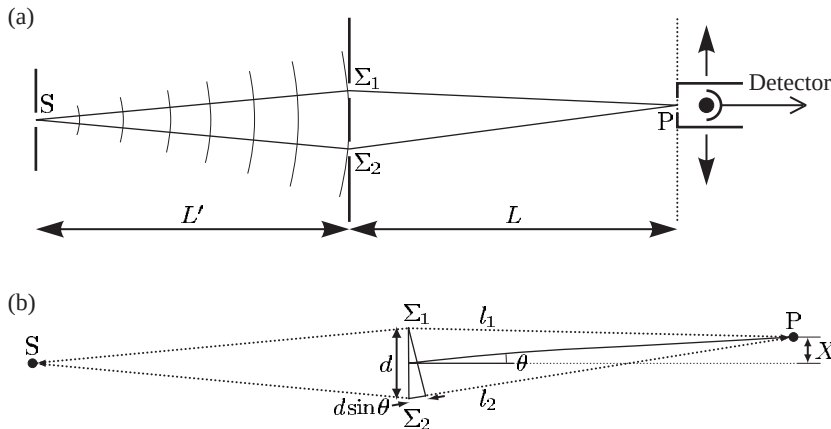


Fig. 11.1 (a) The apparatus for observing interference from double slits. The light diffracted by the source slit propagates through the two slits Σ_1 and Σ_2 and onto the plane P . The interference fringes can be seen with the eye (with the aid of a magnifying eyepiece if necessary), but to further the analogy with atom optics experiments the apparatus is drawn with a detector such as a photodiode or photomultiplier. A narrow slit in front of the detector gives good spatial resolution; this slit and the detector scan across the fringes, as indicated. The light comes from a lamp, or laser; in a matter-wave experiment an atomic oven creates a beam of atoms collimated by the source slit (as shown in Fig. 11.2). (b) The difference in the distance from Σ_1 to P and from Σ_2 to P is $d \sin \theta$, where d is the slit separation. The angle θ and the distance X in the detection plane are related by $X = L \tan \theta$. The transverse distances are drawn greatly exaggerated for clarity; for the typical conditions given in the text the fringes have an angular separation of 2×10^{-3} rad.

⁴ $l_1 = \overline{S\Sigma_1} + \overline{\Sigma_1 P}$ and $l_2 = \overline{S\Sigma_2} + \overline{\Sigma_2 P}$.

we define l_1 as the distance from S to P via Σ_1 , and similarly l_2 for the path through slit Σ_2 , as shown by the dotted lines in Fig. 11.1(b).⁴ Slits of the same size contribute equally to the total amplitude at a point on the plane P:

$$E_P \propto E_0 \left(e^{-i2\pi\ell_1/\lambda} + e^{-i2\pi\ell_2/\lambda} \right). \quad (11.2)$$

The intensity is proportional to the square of this amplitude, $I \propto |E|^2$, so that

$$I = I_0 \cos^2 \left(\frac{\phi}{2} \right). \quad (11.3)$$

Here $\phi = 2\pi(\ell_2 - \ell_1)/\lambda$ is the phase difference between the two arms and I_0 is the maximum intensity. Bright fringes occur at positions in the detection plane where the contributions from the two paths interfere constructively; these correspond to $\phi = n2\pi$, with n an integer, or equivalently

$$\ell_2 - \ell_1 = n\lambda. \quad (11.4)$$

To find the spacing of the fringes in the plane of observation we define the coordinate X measured perpendicular to the long axis of the slits in the plane P. In terms of the small angle defined in Fig. 11.1(b) this becomes

$$X = L \tan \theta. \quad (11.5)$$

A similar small angle approximation allows us to express the path length difference in Fig. 11.1(b) as

$$\ell_2 - \ell_1 = \Delta l + d \sin \theta. \quad (11.6)$$

⁵Elementary treatments often assume that the slits are equidistant from the source ($\Delta l = 0$) so that the phase of the wave is the same at each slit (equivalent to having a plane wavefront before the slits). This assumption simplifies the algebra but it is *not* a necessary condition for Fraunhofer diffraction with light, or with matter waves. (A more detailed description of the assumptions such as $L \gg d$ can be found in textbooks on optics, e.g. Brooker (2003).)

Here $\Delta l = \overline{S\Sigma_1} - \overline{S\Sigma_2}$ is the path difference before the slits.⁵ The path length difference from the two slits of separation d to P is $d \sin \theta$. The last three equations give the spacing of the fringes as

$$\Delta X = \frac{L\lambda}{d}. \quad (11.7)$$

For an experiment with visible light of wavelength $\lambda = 6 \times 10^{-7}$ m, slits of separation $d = 3 \times 10^{-4}$ m and $L = 1$ m, the fringes have a spacing of $\Delta X = 2$ mm and can be clearly seen by eye. The treatment so far assumes a small source slit at S that acts like a point source to illuminate the double slits coherently. The condition for this is that the double slits fall within the angular spread of the light diffracted from the source slit (Brooker 2003). The diffraction from a slit of width w_S has an angular spread $\theta_{\text{diff}} \simeq \lambda/w_S$. Therefore coherent illumination of two slits at a distance L' from this source slit requires $L'\theta_{\text{diff}} \geq d$, or

$$w_S \leq \frac{\lambda L'}{d}. \quad (11.8)$$

For $L' = 0.1$ m and the values of λ and d used previously, we find $w_S \leq 2 \times 10^{-4}$ m. Such slits are made by standard techniques and we know that Young's experiment with light is relatively straightforward to

carry out in the laboratory. Experiments with short-wavelength matter waves require the smallest available structures with slits on the scale of 100 nm.

A double-slit experiment was carried out with a beam of helium atoms in the metastable $1s2s\ ^3S_1$ level that lies 20 eV above the ground state (Carnal and Mlynek 1991). This matter-wave experiment had the same layout as that shown in Fig. 11.1, although with atoms the interferometer has to be set up inside a vacuum chamber. Exercise 11.2 goes through the application of the equations given in this section to the calculation of the slit widths required to observe interference fringes with He^* . Metastable helium is very suitable for this experiment, firstly because it has a fairly long λ_{dB} and secondly when metastable atoms hit a surface they release sufficient energy to eject electrons; counting these charged particles allows the arrival of individual atoms to be detected with high efficiency. Further discussion of double-slit experiments can be found in the quantum mechanics book by Rae (1992); he uses a neutron interference experiment as an example of wave-particle duality.

11.2 A diffraction grating for atoms

Figure 11.2 shows an apparatus with a highly-collimated atom beam of sodium incident upon a transmission grating. The experimenters used a remarkable grating with slits only 50 nm wide, spaced 100 nm apart—equal widths of the bars and the gaps between them. Etching these very thin bars and their delicate support structure represents the state of the art in nano-fabrication. Figure 11.2(b) shows the diffraction pattern obtained with a mixture of sodium atoms and molecules, and Fig. 11.2(c) shows the diffraction of a beam of sodium molecules. The diffraction peaks of Na_2 have about half the spacing of those for the Na atoms, as expected from the de Broglie relation for particles of twice the mass (for similar velocities).

Researchers recently exploited this property of these special gratings to make the first experimental observation of the very weakly-bound state of two helium atoms (Schöllkopf and Toennies 1994). Other methods of detection dissociate the very tenuously-bound He_2 molecule. The gratings work well with helium and atomic beams of inert gases, since they do not clog the slits in the same way as sodium. However, despite the practical difficulties of working with these very fragile structures, a recent experiment used a grating to diffract C_{60} molecules—so-called Buckyballs (Arndt *et al.* 1999, Nairz *et al.* 2003). This demonstration of the wave-like nature of such massive particles prompts the following question: ‘What is the largest object for which such quantum interference can be observed?’⁶ This question relates to Schrödinger’s well-known example of a cat that may be in a superposition of two states (‘alive’ and ‘dead’). To observe interference an object must exist in a superposition of the two states: $|1\rangle$ in which it goes through slit Σ_1 and $|2\rangle$ the state where it goes through slit Σ_2 . Nothing within standard

⁶This means quantum effects in the motion, or external degrees of freedom, and not quantisation of the internal energy levels.

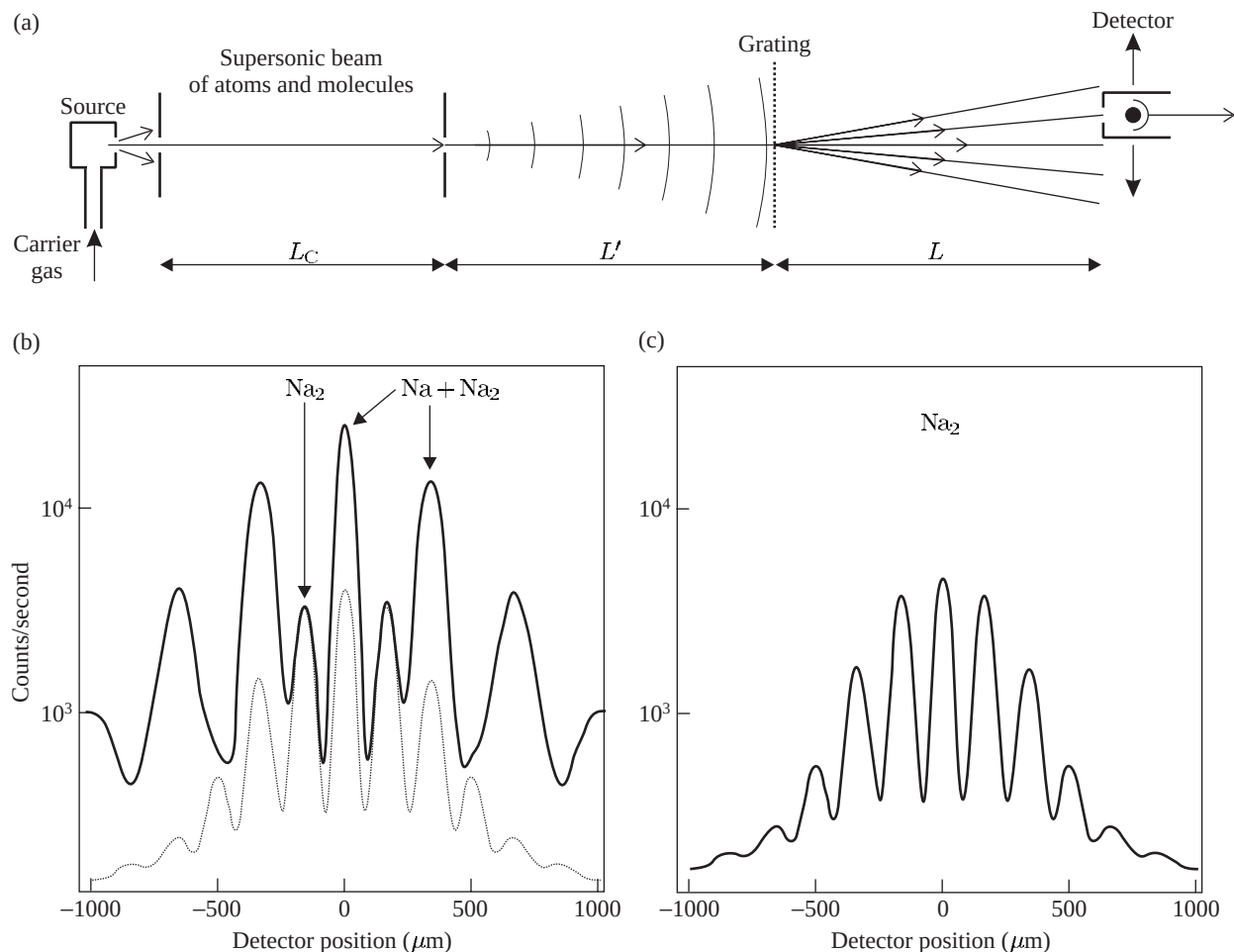


Fig. 11.2 (a) Diffraction of a collimated atomic beam by a grating. To observe the diffraction of matter waves from the grating the source slit must be sufficiently narrow to make the matter waves coherent across several of the slits in the grating. This is the same requirement as described in the previous section for Young's double slits, and Exercise 11.1 looks at the relation of the pattern with multiple slits to that observed with just two slits. For the grating there is an additional requirement that the angular spread of the incident beam must be less than the angle between the diffracted orders, otherwise they cannot be distinguished. In this apparatus the slit widths were about $20\ \mu\text{m}$, the slits in the nano-fabricated grating had a spacing of $100\ \text{nm}$ and all the distances L_C , L' and L were about $1\ \text{m}$. (b) The diffraction of a collimated beam of sodium atoms and molecules by the grating. (c) The diffraction pattern for a beam that contains only Na_2 molecules (this pattern is also shown as a dotted curve in (b)). The peaks for the molecules have half the spacing of those for the atoms as expected for twice the mass—the atoms and molecules have almost the same velocity in the supersonic flow because they are both carried along in a stream of krypton gas that flows through the heated oven containing sodium metal. This carrier gas gives a supersonic beam with a much lower velocity spread $\Delta v/v \simeq 0.03$ than would be obtained from an effusive source of thermal atoms. The sodium atoms were removed from the beam by the resonant radiation pressure from a laser beam (not shown) perpendicular to the supersonic beam—scattering three or more photons was sufficient to deflect atoms out of the beam. From Chapman *et al.* (1995). Copyright 1995 by the American Physical Society.

quantum mechanics tells us that we cannot put something as large as a cat into a quantum superposition, if it is completely isolated from external perturbations. However, the heaviest object that can be used in a practical double-slit experiment in the foreseeable future is far lighter than a cat, but considerably heavier than what has been achieved so far. Continued work on matter-wave interference of larger and larger objects will be of great interest since it probes the boundary between quantum and classical physics.

11.3 The three-grating interferometer

Figure 11.3 shows an arrangement of three diffraction gratings a distance L apart. A highly-collimated beam of sodium atoms propagates through this three-grating interferometer onto a detector for atoms.⁷ Diffraction at the first grating G1 splits the beam—only the zeroth- and first-order diffraction orders (0 and ± 1 orders) have been drawn for simplicity. The second grating G2 gives diffraction through the same angles as G1, so that some of the paths meet up at the plane of the third grating G3, e.g. the 0 and $+1$ orders from G1 are both diffracted by G2 to form the parallelogram ABPC, as shown in Fig. 11.3(a). The detector records the flux of atoms along one of the possible output directions coming from P. This arrangement closely resembles a Mach–Zehnder interferometer for light, with a smaller angle between the two arms because of the achievable grating spacing. For two-beam interference the signal has the same form as eqn 11.3. In these interferometers the sum of the fluxes of the atoms, or light, in the two possible output directions equals a constant, i.e. when a certain phase difference between the arms of the interferometer gives destructive interference at the detector then the flux in the other output direction has a maximum.

⁷The detector for sodium has a hot wire, heated by a current flowing through it, that runs parallel to the slits. The sodium atoms ionize when they hit the hot surface of the wire and the ejected electrons create a measurable current.

11.4 Measurement of rotation

The Mach–Zehnder interferometer for matter waves shown in Fig. 11.3 measures rotation precisely, as explained in this section.⁸ To calculate the phase shift caused by rotation in a simple way we represent the interferometer as a circular loop of radius R , as in Fig. 11.4. The wave travelling at speed v from the point S takes a time $t = \pi R/v$ to propagate around either arm of the interferometer to the point P diametrically opposite S. During this time the system rotates through an angle Ωt , where Ω is the angular frequency of rotation about an axis perpendicular to the plane of the interferometer. Thus the wave going one way round the loop has to travel $\Delta l = 2\Omega R t$ further than the wave in the other arm of the interferometer. This corresponds to $\Delta l/\lambda_{\text{dB}}$ extra wavelengths, or a phase shift of

$$\Delta\phi = \frac{2\pi}{\lambda_{\text{dB}}} \times 2\Omega R \times \frac{\pi R}{v}. \quad (11.9)$$

⁸For light a different configuration called a Sagnac interferometer is generally used to measure rotation but the principles are similar.

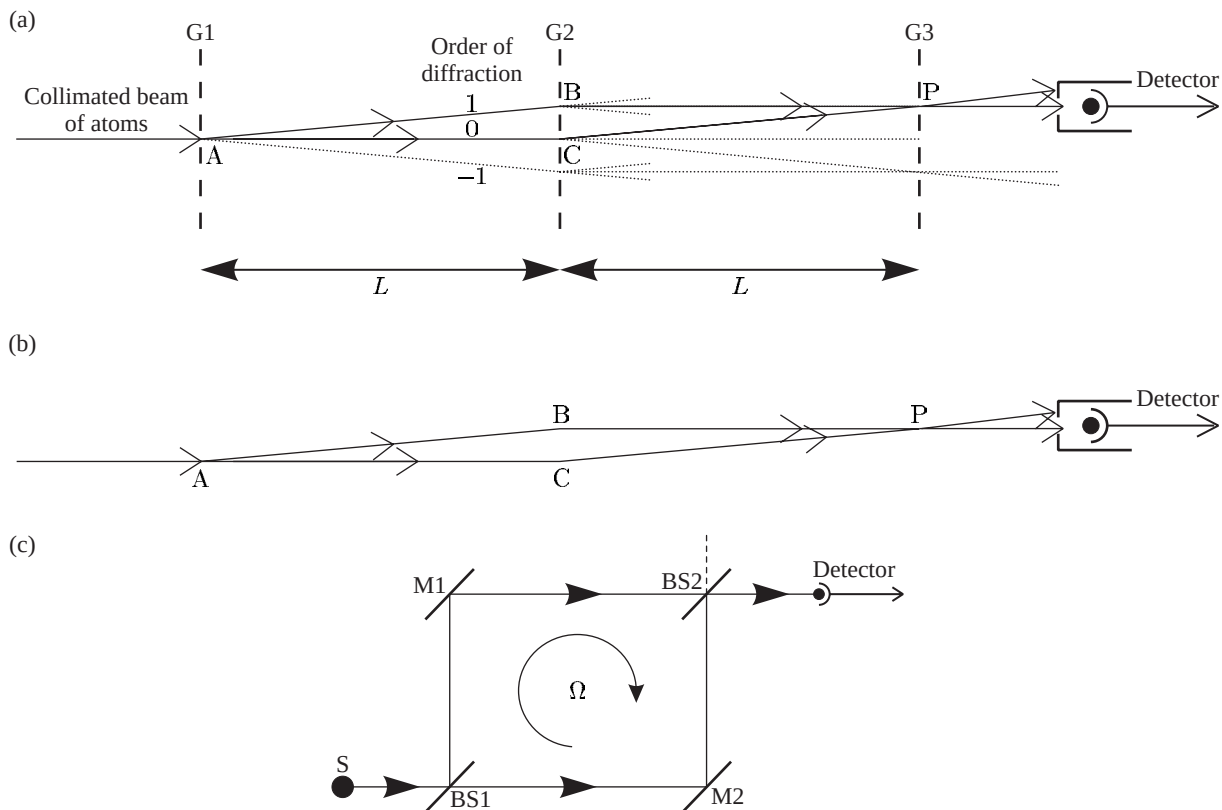


Fig. 11.3 (a) An interferometer formed by three diffraction gratings spaced by a distance L along the atomic beam. A collimated beam of atoms is produced, as shown in Fig. 11.2. Waves diffracted at the first grating G1 split again at G2, so that some of the paths meet at G3. Only the 0 and ± 1 diffraction orders are shown and to further simplify the diagram some of the possible paths between G2 and G3 have not been drawn completely. Contributions to the amplitude at P arrive from A, via either B or C. The detector must be sufficiently far from G3 that it picks up only one of two possible output directions. (The parallelogram ABPC is just one of many closed loops formed by the three gratings; some others are indicated by dotted lines. With the detector at the position shown, the three gratings act as a Mach-Zehnder interferometer, as shown in (b). The diffraction gratings behave both as beam splitters and as deflectors (mirrors) for the matter waves (for small angles). (c) A Mach-Zehnder interferometer for light—the optical system equivalent to the three-grating interferometer. The incident wave hits beam splitter BS1 and the reflected and transmitted amplitudes reflect off mirrors M1 and M2, respectively, so that their paths meet again at BS2. Interference between the two paths leads to a detected intensity $I_D = \frac{1}{2}I_0\{1 + \cos(\phi + \Delta\phi)\}$ (cf. eqn 11.3). The phase ϕ that arises from path length differences and phase shifts on reflection at the mirrors is assumed to be fixed and $\Delta\phi$ represents the extra phase that is measured; e.g. for an interferometer that rotates at angular frequency Ω about an axis perpendicular to the plane of the instrument $\Delta\phi \propto \Omega$, so the instrument measures rotation, as shown in Section 11.4.

The loop has area $A = \pi R^2$, so that

$$\Delta\phi = \frac{4\pi}{\lambda_{dB}v} \times \Omega A. \quad (11.10)$$

A more rigorous derivation, by integration around a closed path, shows that this equation applies for an arbitrary shape, e.g. the square interferometer of Fig. 11.3(c). Comparison of this phase shift for matter waves of velocity v with that for light $\Delta\phi_{\text{light}}$, for an interferometer of the same

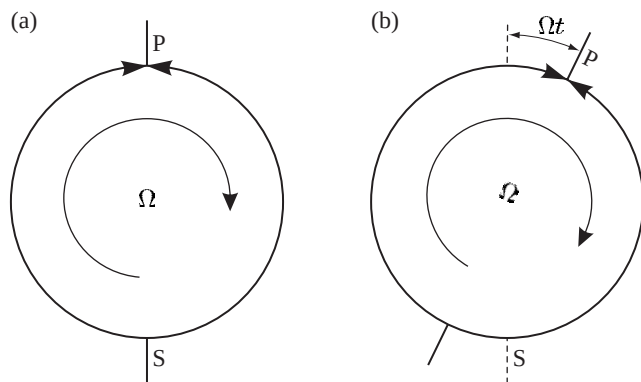


Fig. 11.4 (a) A simplified diagram of an interferometer where the waves propagate from S to P. (b) Rotation at angular frequency Ω about an axis perpendicular to the plane of the interferometer makes one path Ωt longer and the other shorter by the same amount, where t is the time taken for a wave to travel from S to P. This leads to the phase shift in eqn 11.10.

area A , shows that

$$\Delta\phi = \frac{\lambda c}{\lambda_{dB} v} \times \Delta\phi_{\text{light}} = \frac{Mc^2}{\hbar\omega} \times \Delta\phi_{\text{light}}. \quad (11.11)$$

The ratio equals the rest mass of the atom divided by the energy of each photon and has a value of $\Delta\phi/\Delta\phi_{\text{light}} \sim 10^{10}$ for sodium atoms and visible light. This huge ratio suggests that matter-wave interferometers have a great advantage, but at the present time they only achieve comparable results to conventional interferometers with light. Conventional interferometers with light make up the ground by:

- (a) having much larger areas, i.e. a distance between the arms of metres instead of a fraction of a millimetre achieved for matter waves;
- (b) the light goes around the loop many times;⁹
- (c) lasers give a much higher flux than the flux of atoms in a typical atomic beam. For example, in the scheme shown in Fig. 11.2 only a small fraction of the atoms emitted from the source end up in the highly-collimated atomic beam; as a source of matter waves the atomic oven is analogous to an incandescent tungsten light bulb rather than a laser.¹⁰

⁹Laser gyros use high-reflectivity mirrors or optical fibres.

¹⁰The atom interferometer based on Raman transitions (described in Section 11.5.1) does not require such a highly-collimated atomic beam, and that technique has achieved precise measurements of rotation.

11.5 The diffraction of atoms by light

A standing wave of light diffracts matter waves, as illustrated in Fig. 11.5. This corresponds to a role reversal as compared to optics in which matter, in the form of a conventional grating, diffracts light. This section explains how this light field created simply by the retro-reflection of a laser beam from a mirror is used in atom optics. The interaction of atoms with a standing wave leads to a periodic modulation of the atomic energy levels by an amount proportional to the intensity of the light, as explained in Section 9.6.¹¹ The light shift of the atomic energy levels in the standing wave introduces a phase modulation of the matter waves. An atomic wavepacket $\psi(x, z, t)$ becomes $\psi(x, z, t)e^{i\Delta\phi(x)}$ immediately after passing through the standing wave; it is assumed that

¹¹The laser has a frequency ω sufficiently far from the atom's transition frequency ω_0 that spontaneous emission has a negligible effect, yet close enough for the atoms to have a significant interaction with the light.

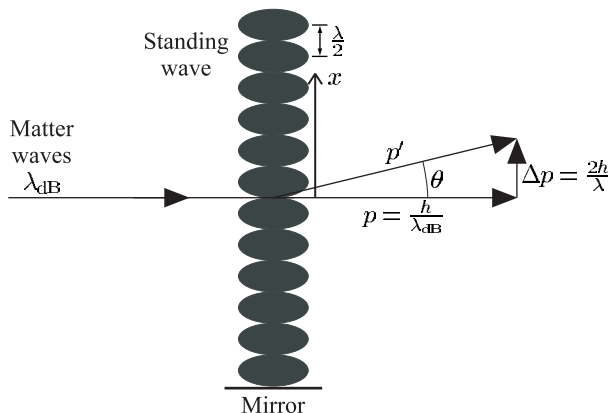


Fig. 11.5 The diffraction of atoms by a standing wave light field. The angle of the first order of diffraction θ is related to the grating period d by $d \sin \theta = \lambda_{\text{dB}}$. For the standing wave $d = \lambda/2$, so $\sin \theta = 2\lambda_{\text{dB}}/\lambda$. This diffraction can be regarded as a scattering process where an atom of momentum p receives an impulse that gives it transverse momentum Δp and deflects it by an angle θ given by $\tan \theta = \Delta p/p$.

$\psi(x, z, t)$ changes smoothly over a length scale much greater than $\lambda/2$. This phase modulation has a spatial period of $\lambda/2$, where λ is the wavelength of the *light* not the matter waves. The matter waves accumulate an additional phase $\Delta\phi(x) = \phi_0 \cos^2(2\pi x/\lambda)$ from the light shift. This phase grating diffracts the matter waves at angles determined by

$$d \sin \theta = n \lambda_{\text{dB}}, \quad (11.12)$$

where $d = \lambda/2$, n is an integer and λ_{dB} gives the matter wavelength. Gratings with the same spacing d diffract waves by the same angles θ whether they work by phase or amplitude modulation of the incident wave,¹² although not with the same relative intensities in different orders. Thus there is no fundamental difference between nano-fabricated absorption gratings and the use of standing waves. Interferometers that use three standing waves in the same arrangement as in Fig. 11.3 have similar properties to an instrument with three gratings etched from solid material. The mirrors that retro-reflect the light to form the standing waves must be mounted rigidly so that vibrations do not wash out the interference fringes (similarly, the nano-fabricated gratings must be held very stable). Standing waves of visible light give diffraction angles about three times less than those from the best nano-fabricated gratings; however, such gratings transmit all the atoms, whereas nano-fabricated gratings transmit much less than the 50% that might be expected for bars that have a width equal to the gaps between them—the very thin bars require an elaborate support structure of cross-bars that reduces the open area. Also, material gratings eventually get clogged up when used with alkali metals.

The diffraction of the matter waves by a standing wave has an alternative physical interpretation in terms of the scattering of light. The diffraction condition in eqn 11.12 can be written as $\tan \theta \simeq n \hbar G/p$ (taking $\tan \theta \simeq \sin \theta$ for small angles), where p is the longitudinal momentum of the atoms, $G = 4\pi/\lambda$ is the characteristic wavevector of a structure with spacing $\lambda/2$ (i.e. the grating) and $\hbar G = 2\hbar/\lambda$ equals the momentum of two photons. Thus, in the diffraction from the standing wave, atoms

¹²This is obvious from a treatment of diffraction by Fourier transforms (Brooker 2003).

receive a transverse kick from $2n$ photons, as illustrated in Fig. 11.5, e.g. first order arises from absorption from one of the counter-propagating beams and stimulated emission into the other beam. This coherent process, with no spontaneous emission, has some similarities with the Raman transition shown in Fig. 11.6.¹³

¹³Scattering in a standing wave changes the atomic momentum (external state) but not the internal state.

11.5.1 Interferometry with Raman transitions

The description of the diffraction of two-level atoms by a standing wave, in the previous section, as a coherent scattering process that imparts twice the photon momentum $2\hbar/\lambda$, or multiples thereof, to the atom has links with the very powerful method for manipulating the atom's momentum by Raman transitions shown in Fig. 11.6. Two laser beams at frequencies ω_{L1} and ω_{L2} drive a coherent Raman transition between states $|1\rangle$ and $|2\rangle$ when

$$\hbar(\omega_{L1} - \omega_{L2}) = E_2 - E_1. \quad (11.13)$$

No population goes into the intermediate state $|i\rangle$ in this coherent transition because neither of the two beams excites a single-photon transition (see Appendix E). The Raman transition couples states $|1\rangle$ and $|2\rangle$ and drives Rabi oscillations between them, e.g. when the atom starts in either $|1\rangle$ or $|2\rangle$, a $\pi/2$ -pulse creates a superposition of $|1\rangle$ and $|2\rangle$ with equal amplitudes. Raman laser beams propagating in opposite directions (as in Fig. 11.6) change the atom's momentum during the transition.¹⁴ The absorption of a photon of wavevector $\mathbf{k}_1$ and the stimulated emission of one in the opposite direction $\mathbf{k}_2 \simeq -\mathbf{k}_1$ gives the atom two recoil kicks in the same direction. This process couples the state $|1, p\rangle$ to $|2, p + 2\hbar k\rangle$. The bra(c)ket notation denotes the $|\text{internal state, momentum}\rangle$ of the atoms. The Raman resonance condition in eqn 11.13 depends sensitively on the atom's velocity v for counter-propagating beams and this provides the basis for the Raman cooling of atoms (Section 9.8). For interferometry this velocity selectivity is a complicating factor and we shall assume that the Raman pulses are sufficiently short¹⁵ to drive transitions over the whole range of velocity components along the laser beam.

¹⁴Raman laser beams travelling in the same direction have the same effect as direct coupling between $|1\rangle$ and $|2\rangle$ by microwaves.

¹⁵According to the condition in eqn 9.58.

Figure 11.7 shows a complete Raman interferometer where the atoms start in $|1, p\rangle$ and travel through three Raman interaction regions. In

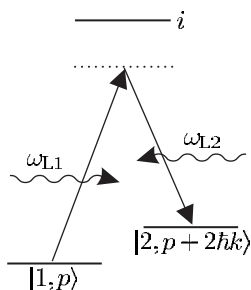


Fig. 11.6 A Raman transition with two laser beams of frequencies ω_{L1} and ω_{L2} that propagate in opposite directions. Equation 11.13 gives the resonance condition, ignoring the effects of the atom's motion (Doppler shift). The Raman process couples $|1, p\rangle$ and $|2, p + 2\hbar k\rangle$ so that an atom in a Raman interferometer has a wavefunction of the form $\psi = A|1, p\rangle + B|2, p + 2\hbar k\rangle$ (usually with either $B = 0$ or $A = 0$ initially).

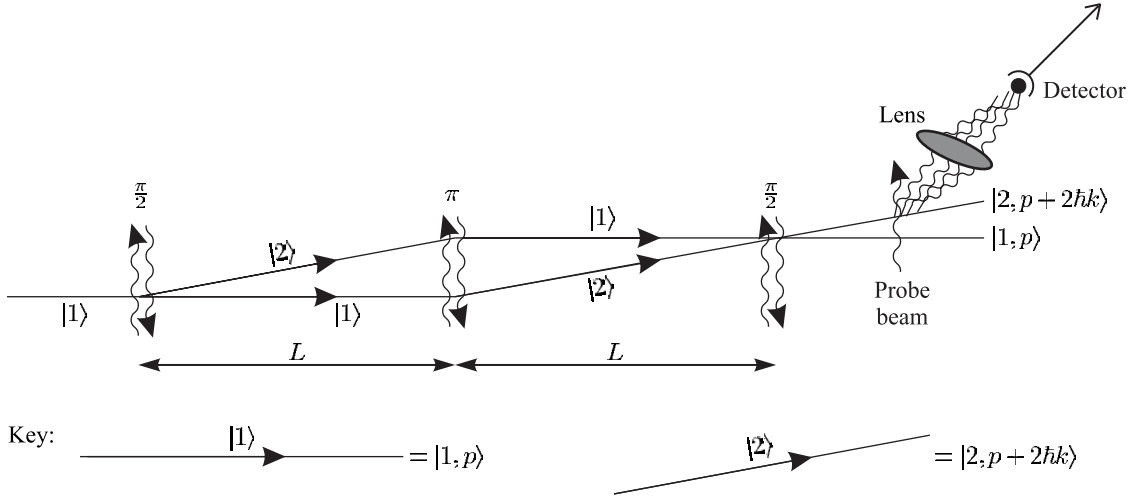


Fig. 11.7 An interferometer formed by Raman transitions. As atoms traverse the three interaction regions they experience a $\pi/2$ – π – $\pi/2$ sequence of Raman pulses that split, deflect and recombine the atomic wavepackets. Each interaction region has two counter-propagating beams at frequencies ω_{L1} and ω_{L2} , as in Fig. 11.6. In this Mach–Zehnder interferometer the eigenstates with transverse momentum p and $p + 2\hbar k$ are associated with the different internal atomic states $|1\rangle$ and $|2\rangle$, respectively; as indicated at the bottom of the figure, $|1\rangle$ is associated with $p \simeq 0$ (horizontal line) and $|2\rangle$ with $p \simeq 2\hbar k$ (slanted line). Therefore at the output it is only necessary to measure the internal state of the atoms, e.g. by exciting a transition from $|2\rangle$ and detecting the fluorescence, rather than allowing beams with different momenta to become spatially separated, as in Fig. 11.3. (The separation of the paths has been exaggerated for clarity.)

the first interaction the atom experiences a $\pi/2$ -pulse that puts it into the superposition

$$|\psi\rangle = \{ |1, p\rangle + e^{i\phi_1} |2, p + 2\hbar k\rangle \}. \quad (11.14)$$

¹⁶This phase, and the other similar phases that arise at each interaction, lead to an offset in the final output which is not important. But these phases must remain constant in time, otherwise the interference ‘washes out’.

The phase factor depends on the relative phase of the two laser beams.¹⁶ These two states separate, as shown in Fig. 11.7, and this first region corresponds to a beam splitter for matter waves. After a free-flight through a distance L the atom enters the middle interaction region where it undergoes a π -pulse that acts on both arms of the interferometer to swap the states $|1, p\rangle \leftrightarrow |2, p + 2\hbar k\rangle$. (In this apparatus the transit time for the atom to pass through the laser beams determines the duration of the Raman interaction.) The paths come back together after a further distance L and the final $\pi/2$ -pulse acts as the beam splitter that mixes the wavepackets to give interference. The complete $\pi/2$ – π – $\pi/2$ sequence gives a Mach–Zehnder interferometer. A comparison of Figs 11.3 and 11.7 shows that the Raman scheme resembles the Mach–Zehnder interferometer more closely than the three-grating device; the Raman scheme does not direct any amplitude in unwanted directions and the middle interaction region in the Raman interferometer acts just like a mirror to change the direction (transverse momentum) of both paths through a small angle.¹⁷

¹⁷In a three-grating interferometer only a fraction of the amplitude goes in the required direction. Note, however, that the simple treatment of standing waves assumed the ‘thin’ grating approximation, but often the interaction between the matter waves and light takes place over a sufficiently long distance that Bragg diffraction occurs (as in crystals).

A Raman pulse and standing light wave give the same opening angle between the arms for a given wavelength of laser light and both

schemes use light whose frequency is detuned from the atomic transition to avoid spontaneous emission. A crucial difference between these methods arises in the detection. The three-grating apparatus, with standing waves or nano-fabricated structures, distinguishes the two outputs by their different directions. Therefore the three-grating devices require a highly-collimated atomic beam at the input whose angular divergence is less than the angle between the two output directions θ_{diff} . The output channels of the Raman scheme are the two different states $|1, p\rangle$ and $|2, p + 2\hbar k\rangle$, as shown in Fig. 11.7; thus experiments only need to determine the final state of the atom, e.g. using a laser beam that excites a transition from $|2\rangle$ to another state that gives fluorescence for atoms in state $|2\rangle$ but not for those in $|1\rangle$.¹⁸ This means that Raman interferometers use more of the atoms from a given source because they do not need to have tight collimation.¹⁹ Although the flux of atoms does not affect the size of the phase shift given by eqn 11.10, the strength of the measured signal determines how precisely that phase shift can be measured, i.e. the interferometer measures a smaller fraction of a fringe if the signal-to-noise ratio is higher.²⁰ Thus the type of Raman interferometer shown in Fig. 11.7 measures rotation more precisely than a three-grating device.

¹⁸As in the atomic fountain described in Section 9.9.

¹⁹The creation of the two Raman beams with a well-defined frequency difference, and other technical details, are described in Section 9.8.

²⁰This argument assumes that it is purely statistical fluctuations (noise) that limit the precision, not systematic shifts.

11.6 Conclusions

Matter-wave interferometers for atoms are a modern use of the old idea of wave–particle duality and in recent years these devices have achieved a precision comparable to the best optical instruments for measuring rotation and gravitational acceleration. We have seen examples of experiments that are direct analogues of those carried out with light, and also the Raman technique for manipulating the momentum of atoms through their interaction with laser light, as in laser cooling. Laser cooling of the atom’s longitudinal velocity, however, only gives an advantage in certain cases (see the section on further reading).²¹ Similarly, the high-coherence beams, or atom lasers, made from Bose condensates do not necessarily improve matter-wave devices—in contrast to the almost universal use of lasers in optical interferometers. Partly, this arises because of the interactions between the atoms themselves, as discussed in the derivation of the nonlinear Schrödinger equation in Chapter 10, which lead to phase shifts that depend on the atomic density. So far interferometry experiments that use BEC have been performed to find out more about the condensate itself, rather than as instruments for precision measurement of physical quantities. The interaction of atoms with the periodic potential produced by a standing wave gives a lot of interesting physics, in addition to the diffraction described here, and we have only scratched the surface of atom optics.

²¹The Ramsey fringes produced by atomic fountain clocks arise from interference of the internal (hyperfine) states of atoms, but in this chapter the ‘atom interferometer’ has been reserved for cases where there is spatial separation between the two arms.

Further reading

²²This journal is a useful source of similar articles.

The review in *Contemporary physics*²² by Godun *et al.* (2001) surveys the field of atom interferometry at a level suitable for undergraduates, including important applications, such as the precision measurement of gravitational acceleration g , that have not been included here. The monograph *Atom interferometry* edited by Berman (1997) is a rich source of information on this subject.

Exercises

(11.1) *Comparison of double- and multiple-slit diffraction*

- Explain in simple physical terms why the diffraction orders of a grating occur at the same angles as the constructive interference between a pair of slits with the same spacing as those in the grating.
- Monochromatic light passes through a transmission diffraction grating. Initially most of the grating is covered with opaque sheets of material so that the light illuminates only two adjacent slits in the middle of the grating. The grating is gradually uncovered until finally light falls on the whole grating. Describe how the intensity, spacing and shape of the observed far-field diffraction changes?

(11.2) *Young's slits with atoms*

- Calculate λ_{dB} for metastable helium atoms from a source at 80 K.
- Find the source slit width w_{S} such that the diffracted wave spreads out to coherently illuminate two slits separated by $d = 8 \mu\text{m}$ when $L' = 0.6 \text{ m}$ in Fig. 11.1 (the conditions for the experiment of Carnal and Mlynek (1991)).

(11.3) *Measurement of the van der Waals interaction with a nano-fabricated grating*

Diffraction by a grating with slits of width a and spacing d gives an intensity distribution of²³

$$I = I_0 \left(\frac{\sin(Nud/2)}{\sin(ud/2)} \right)^2 \left(\frac{\sin(ua/2)}{ua/2} \right)^2.$$

Here $u = 2\pi \sin \theta / \lambda_{\text{dB}}$ and the angle is defined in Fig. 11.1. All of the parts of this exercise refer to a grating with $d = 2a = 100 \text{ nm}$.

- Sketch the intensity distribution for $0 \leq u \leq 10\pi/d$.
- What is the intensity of the second order?
- An experimental observation of the diffraction of rare gas atoms from the grating found that the intensity of the second order is $0.003 I_0$ for helium and $0.05 I_0$ for krypton. The difference in these values was ascribed to the van der Waals force, which is strongest for large atoms. Therefore a krypton atom on a trajectory that goes close to the sides of the slit feels a force that deflects it through a large angle or causes it to crash into the grating. These processes effectively reduce the slit width from a to $a - 2r$, where r is the typical van der Waals range. Estimate r for krypton atoms.²⁴

Web site:

<http://www.physics.ox.ac.uk/users/foot>

This site has answers to some of the exercises, corrections and other supplementary information.

²³Brooker (2003).

²⁴Based on the experiment of Grisenti *et al.* (1999).